

Thesis projects, 2026-2027

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1. Developer networks

This study investigates **how the introduction of artificial intelligence (AI) tools and remote work (work-from-home, WFH) practices influence collaboration and communication in software development teams.**

We focus particularly on the evolving structure of interactions between senior and junior developers and aim to understand how these dynamics affect knowledge exchange and organizational learning.



Repository metadata - GitHub

e.g. commits, pull requests, code ownership from GitHub



Task management data

e.g. ticket creation, assignments, comments from Linear

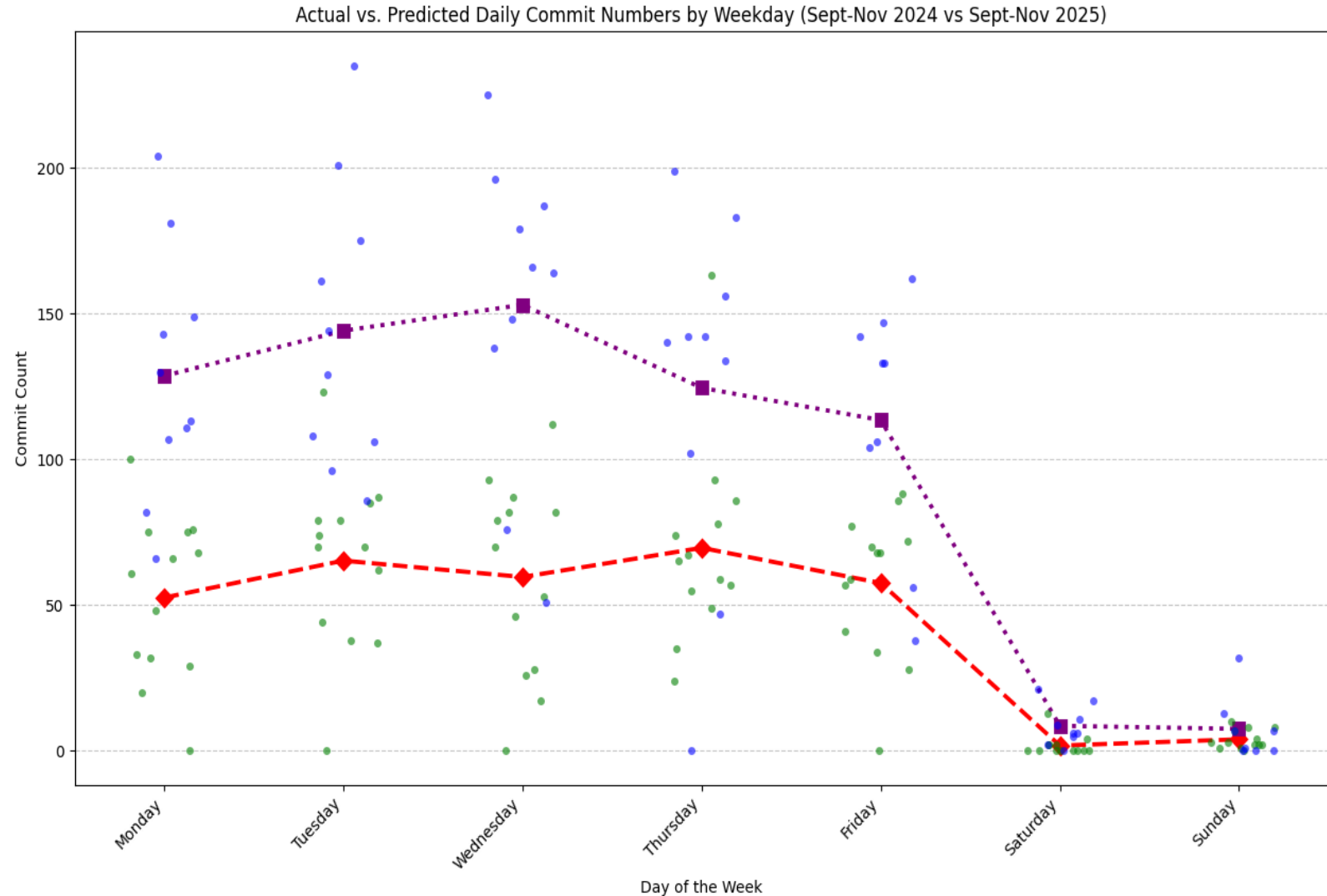


Communication logs

e.g. message frequency, channels, mentions from Slack

The policy changed

- The mandatory office days are extended
- Initial Tuesday and Thursday mandatory office days
- After Sept 2025 Monday became also mandatory office day
- Compare 2024 & 2025
- Wednesday activity data diverge
- Other days moves together

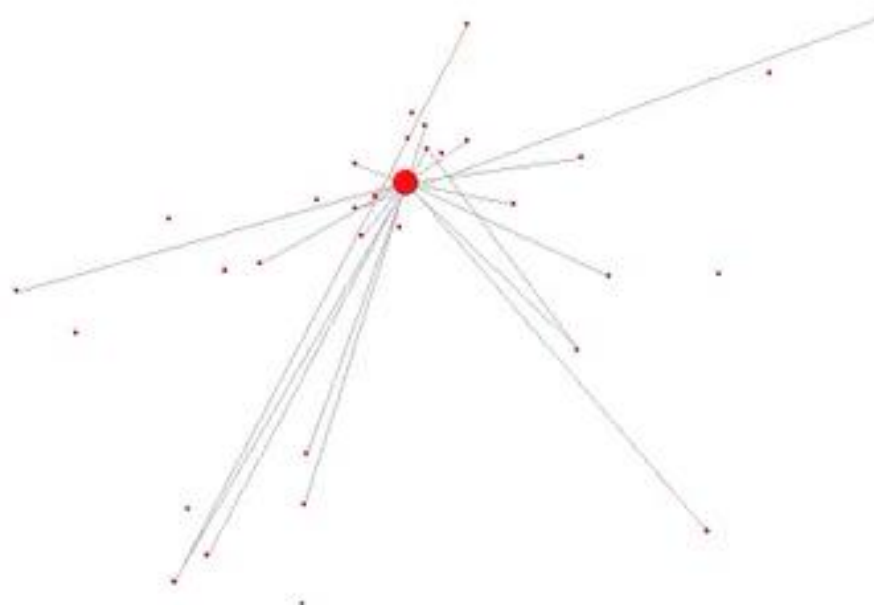
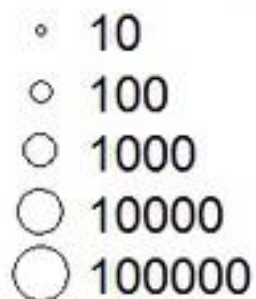


2. Spatial adoption and churn



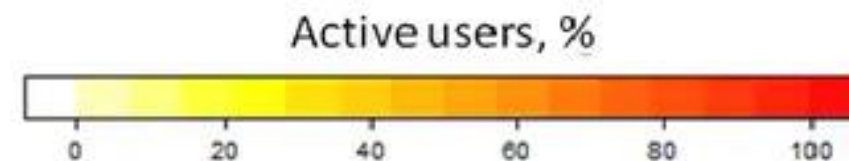
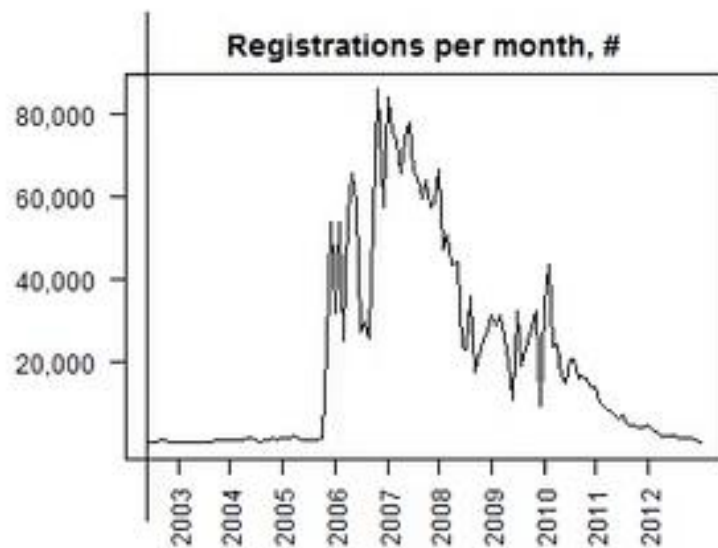
Full life-cycle of a social media platform

Registered users, #

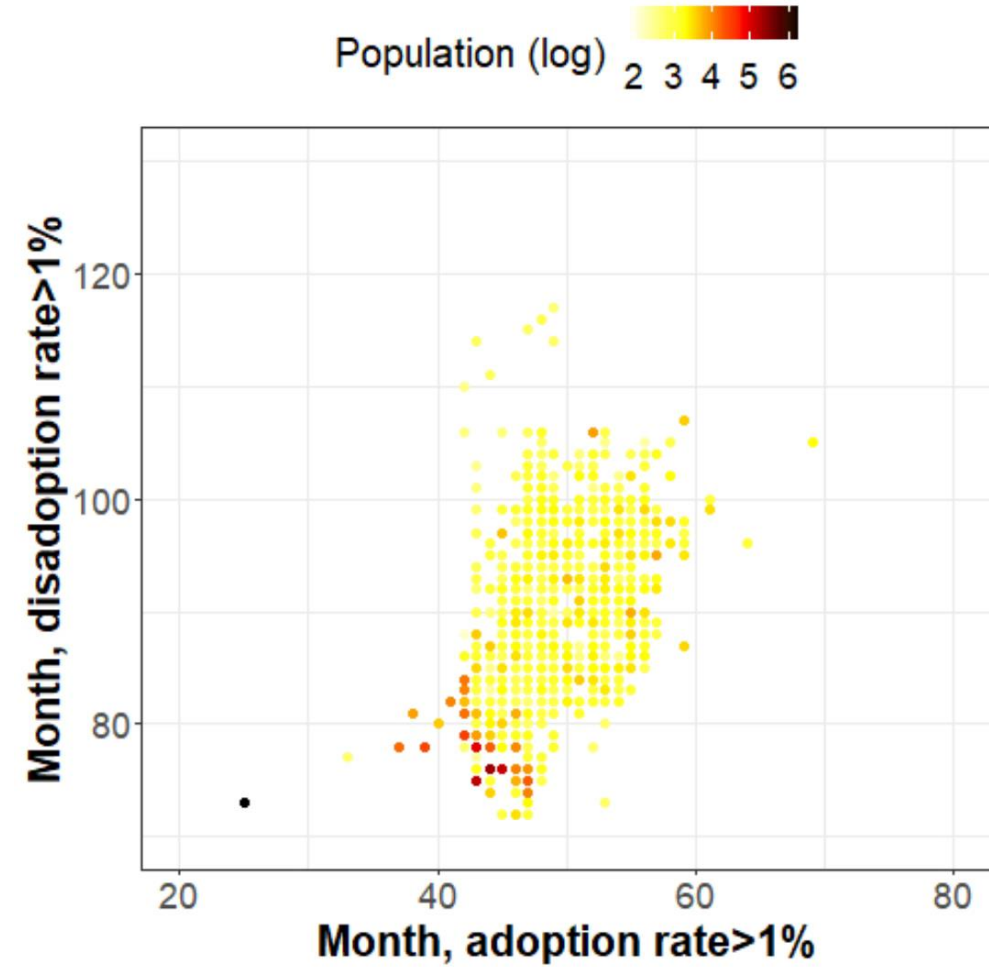
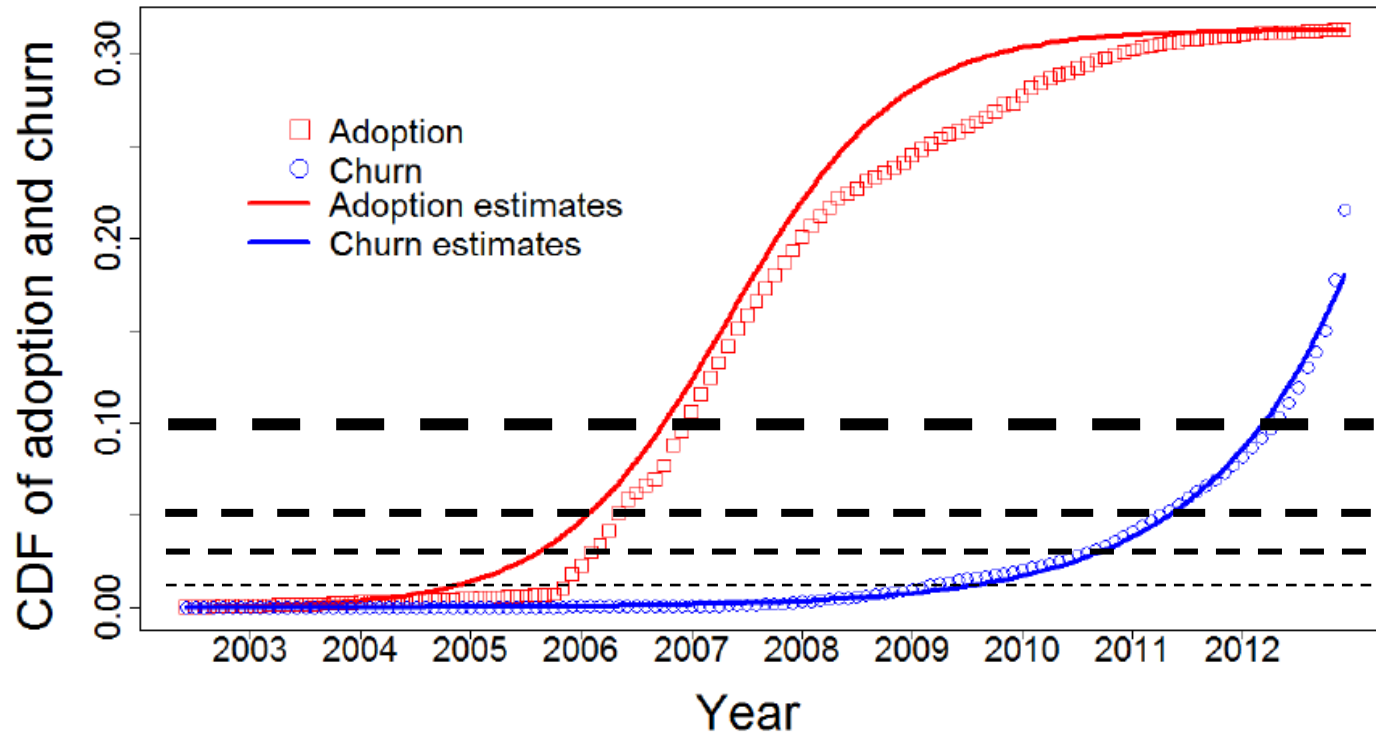


2.8 M Users

- Location (2557 towns),
- Date of registration,
- Date of last login,
- Connections (time-stamp),
- ID of inviters



Those places churn early that adopt early

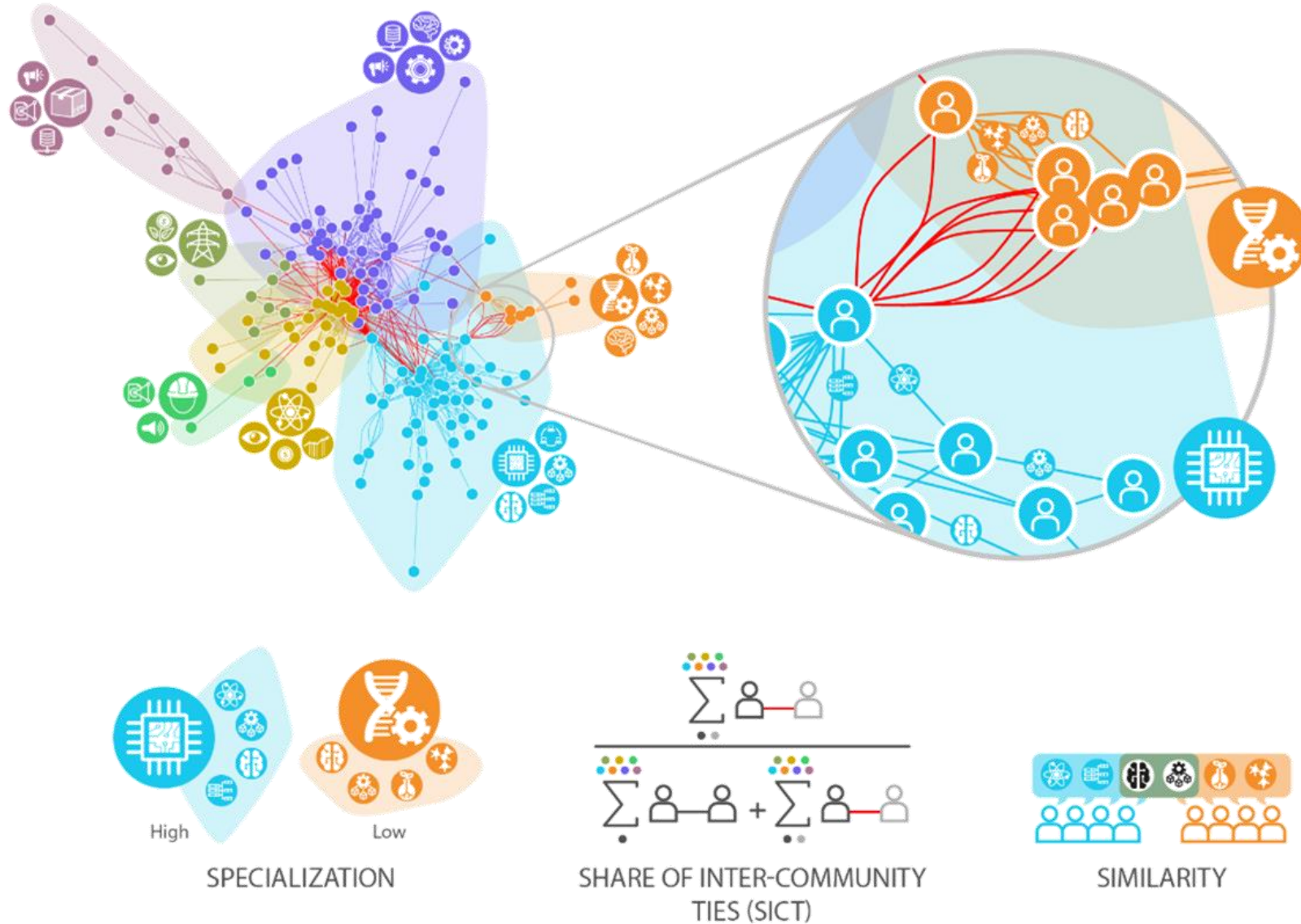


Further steps: generalization and synergies

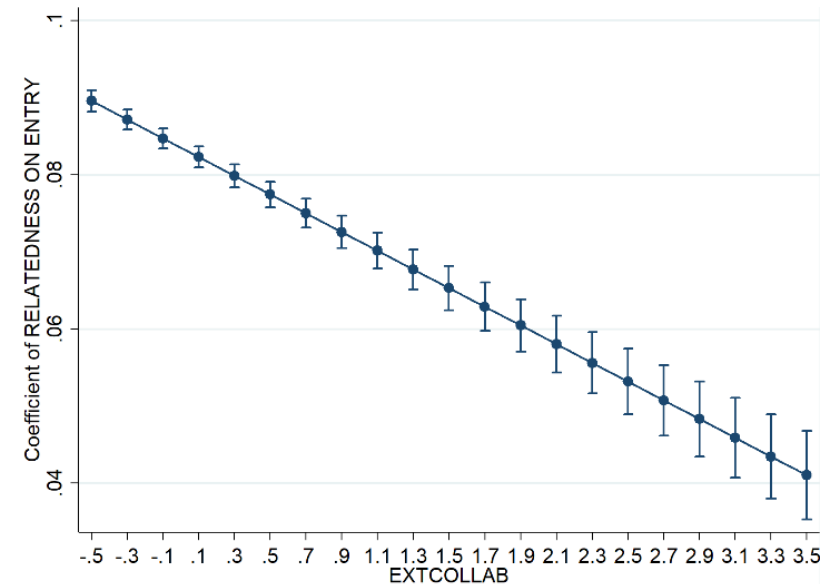
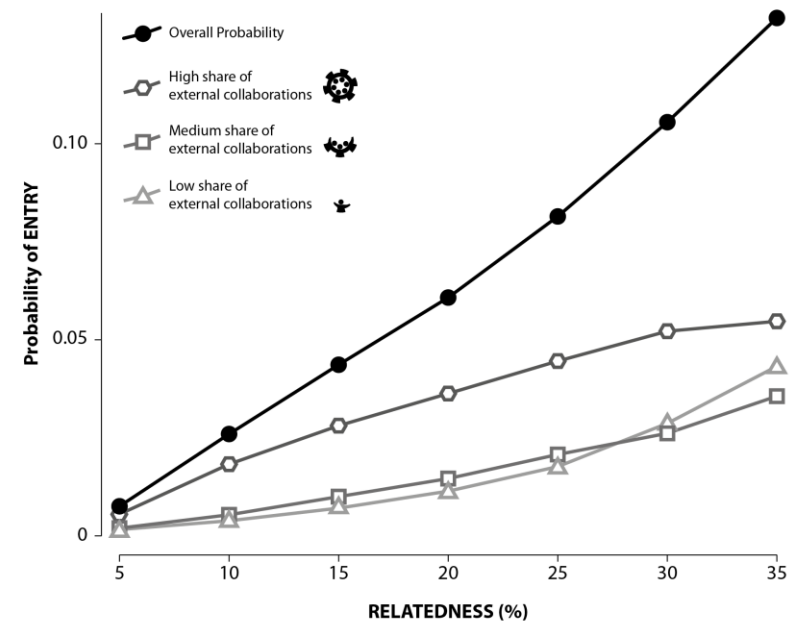
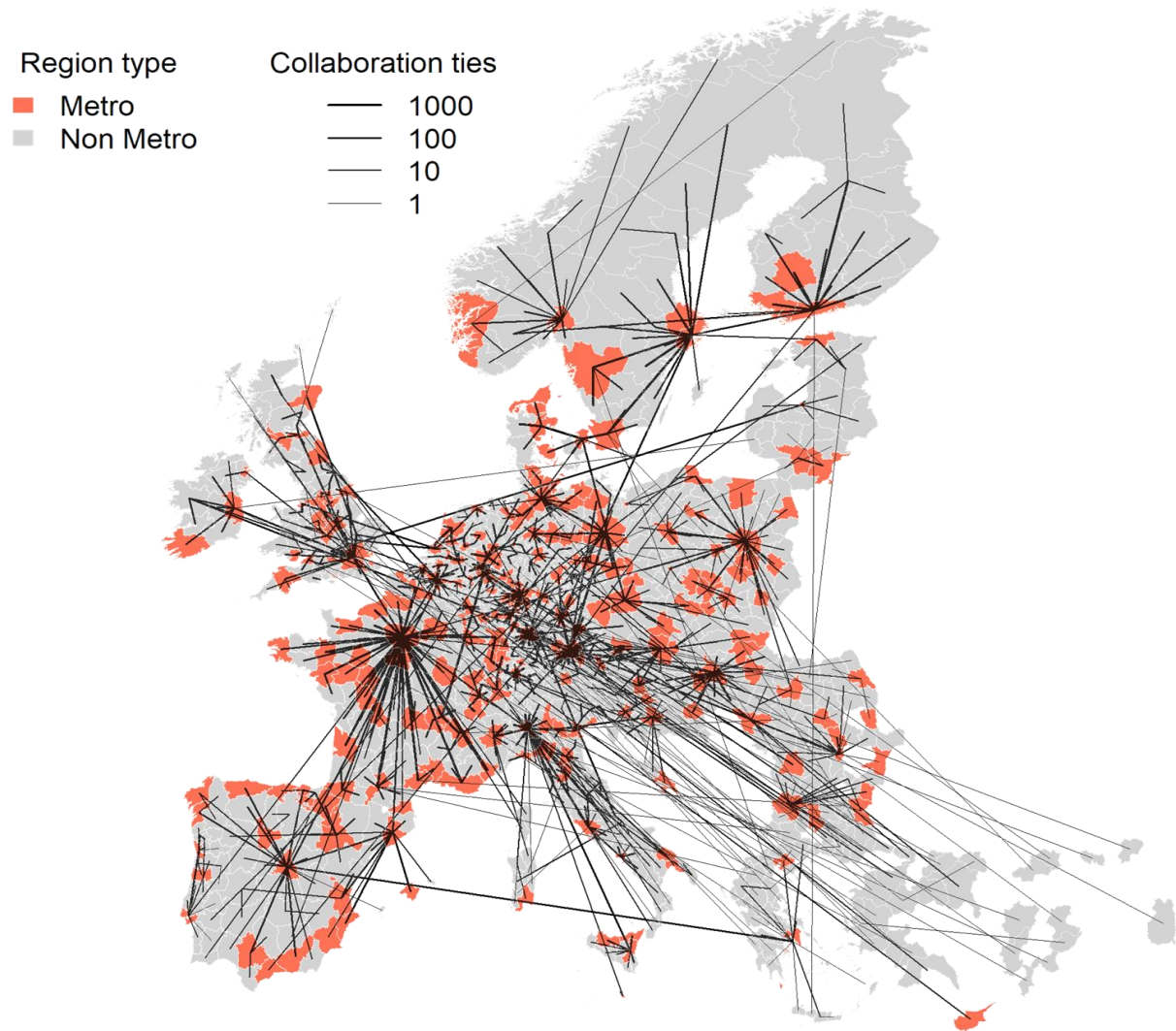
Different contexts

- Web technologies
- Web search (Google Trends)
- Technologies from patent (and trademark) data
- Scientific terms, concepts, keywords in publications
- Topics and expressions in news

3. Novelty in scientific and innovative collaboration



New specializations in the region



Thank you!



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